HORMONES OF THE GASTROINTESTINAL TRACT

The word hormone is derived from the Greek word 'HORMEIN', which means to execute or to arouse. It is secreted from the ductless glands, goes to the circulation and acts on their target cells, where they elicit physiologic, morphologic and biochemical responses.

THE MAIN FUNCTION OF THE GASTROINTESTINAL TRACT ARE –

- Digestion
- Absorption
- Nutrition

CLASSIFICATION

- Gastrin family
  - Gastrin
  - Cholecystokinin
- Secretin family
  - Secretin
  - Gastric inhibitory peptide
  - Vasoactive intestinal peptide
  - Glucagon
  - Insulin
  - Leptin

GASTRIN

- Secreted by 'G' cells, located in the pyloric region of the stomach
- It is flasked shaped, secreted in inactive form, converted to active form by HCL.
- Three types are there – G-34, G-17, G-14
- Its plasma half-life is 2-3 minutes
- Inactivated mainly in the kidney and the small intestine

  FUNCTIONS –
  - Stimulates HCL secretion
  - Increases intestinal and gastric motility
  - Increases pancreatic secretion of INSULIN and GLUCAGON
  - TROPHIC ACTION i.e. maintain the health of gastric mucous membrane.

  FACTORS AFFECTING SECRETIONS
  - Vagal stimulation
  - Distension of the pyloric antrum
  - Products of protein digestion
  - Calcium and epinephrine

  FACTORS INHIBITING SECRETION
  - Low pH of gastric juice (less than 3.)
  - Somatostatin released by D cells.
  - Others – SECRETIN, GIP, VIP, GLUCAGON, CALCITONIN

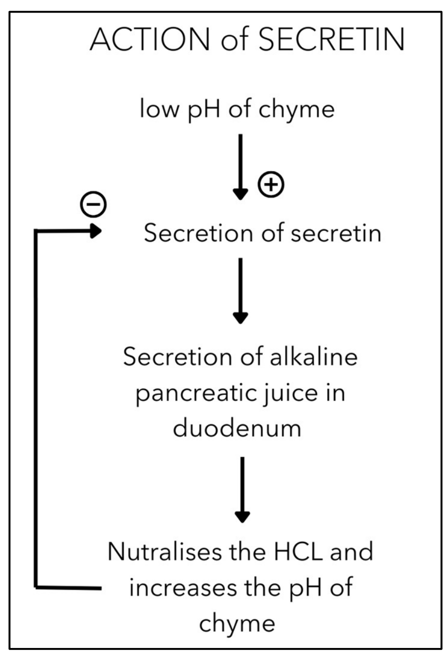
APPLIED
- Zollinger ellison syndrome
- Some pancreatic tumors of delta cells, secrete excess HCL and causes peptic ulceration
- Gastrinoma

CHOLECYSTOKININ PANCREOZYMIN
- It is secreted from the I cells of the duodenum and jejunum.
- It is a polypeptide containing 33 amino acids

ACTIONS
- Contraction of gall bladder to release bile.
- Potentiates the secretion of 'SECRETIN' to produce more alkaline juice.
- Inhibits the GASTRIC MOTILITY.
- Increases the motility of small and large gut.
- Trophic action
- It is also involved in regulation of food intake

APPLIED
- Acute pancreatitis
- Chronic pancreatitis

SECRETIN
- It was the first hormone to be discovered in 1902, by BAYLISS and STARLING.
- Secreted by 'S' cells of duodenum and jejunum.
- It is a polypeptide with 27 amino acids.
- Also stimulates 'BILE' secretion.
- Potentiates the effect of CCK on pancreas.
- Causes contraction of the pyloric sphincter delaying gastric emptying and preventing duodenal reflux.



GLUCAGON
- It is secreted by alpha cells of the 'ISLETS OF LANGERHANS' when the blood glucose level falls.
- It is a polypeptide containing 29 amino acids
- It is also secreted in the 'L' cells in the lower and also in the brain.

INCREASE SECRETION
- Hypoglycemia
- Stimulation of the sympathetic nerves to the pancreas
- Vagal stimulation
- Protein meal

- During starvation
- CCK
- Gastrin

DECREASE SECRETION

- Secretin
- Rise in plasma glucose level

ACTIONS

ON CARBOHYDRATE METABOLISM

- Increase glycogenolysis in the liver.
- Increase gluconeogenesis FROM lactate, pyruvate, glycerol, amino acids

INSULIN

- It is a peptide hormone, released by beta cells of the pancreas
- Discovery by BENTING and BEST. In 1922.
- It is secreted when the blood glucose exceeds the normal level i.e. 70-110 mg/dl
- Insulin receptors are present in most cells, [INSULIN DEPENDENT]
- Cells like – BRAIN, KIDNEY, LINING OF G.I.T, RBC CELLS [INSULIN INDEPENDENT] . no insulin receptors can absorb GLUCOSE without insulin.
- EFFECT OF INSULIN ON VARIOUS TISSUE
    - Adipose tissue
    - Muscle
    - Liver
    - General
- APPLIED
    - Diabetes mellitus
        - It is a syndrome of impaired carbohydrate, fat and protein metabolism caused by either lack of insulin secretion or sensitivity of the tissue to insulin.
        - Types – TYPE -1 , TYPE- 2

THE STOMACH

- The stomach is dilated part of the alimentary canal that between oesophagus and small intestine.
- It is a muscular sac
- It is J shaped.

THE STOMACH IS DIVIDED INTO FOUR REGIONS

1. The **cardia**, which surrounds the opening of the esophagus into the stomach.
2. The **fundus of stomach**, which is the area above the level of the cardial orifice.
3. The **body of stomach**, which is the largest region of the stomach.

4. The **pyloric part**, which is divided into the **pyloric antrum** and **pyloric canal** and is the distal end of the stomach.

LINING OF GASTRIC MUCOSA

- Inner surface of stomach is highly folded into rugae.
- Gastric mucosa has gastric pits.
- Cells that lines the folds deeper in the mucosa are the gastric glands
- Gastric glands are situated within the mucous membrane in groups of 2-7 which together communicate with the gastric lumen through gastric pits.

GASTRIC GLANDS ARE MAINLY THREE TYPES

1. Main gastric glands – found in body and fundus of stomach. Cells of these glands are 2 types
    a. Chief/peptic cells – secrete pepsinogen
    b. Parietal/oxyntic cells – secrete hydrochloric acid
2. Cardiac tubular glands – secrete soluble mucous
3. Antral gland – contain 2 types of cells
    a. Mucus cells – secrete soluble mucus
    b. G cells – secrete Gastrin

CHIEF CELLS/ PEPTIC CELLS
Situated in the wall of the glands near the base, Secretes pepsinogen.
PEPSINOGEN –
The functions of pepsinogen are –

a. Pepsinogen digest proteins to polypeptides by hydrolysing peptide bonds. pH for this action requires <4.0
b. Pepsin acts on water soluble easeinogen (milk protein) to form soluble casein. This combines with calcium salts to form insoluble calcium caseinate which is digested enzymatically.

PARIETAL CELLS/OXYNTIC CELLS
It is found in the body of the stomach. They secrete HCL as well as INTRINSIC FACTOR (IF). pH of this fluid is 1.0
FUNCTION OF HCL

1. Provides H+ concentration necessary for pepsin activity.
2. Activates pepsinogen to pepsin.
3. Kills ingested bacteria
4. Helps in iron absorption by converting ferric to ferrous form.

*The parietal cells also secretes IF which combines with vitamin B12 and helps in absorption of the latter from the small intestine.

CARDIAC TUBULAR GLAND
Secretes soluble mucus.

PYLORIC GLAND/ANTRAL GLANDS
They are found in the mucosa between the level of incisura angularis on lesser curvature and pylorus. Mucus cells secrete mucus. Deeper part of the pyloric gland have G cells which secrete **GASTRIN**

FUNCTION OF STOMACH
1. Storage
2. Digestive
3. Function of HCL
4. IF (intrinsic factor) : requires for absorption of vit B12. Decrease absorption produces megaloblastic anemia.
5. Food released at a controlled rate into the duodenum to provide proper time for digestion and absorption by small intestine.

COMPOSITION OF GASTRIC JUICE

Daily secretion	:	2.5 – 3 liters/day, isotonic
Ph	:	1 – 2 (acidic)
Water	:	99.45%
Solids	:	0.55%

SOLIDS
A. Electrolytes
   a. Cation : $Na+$; $K+$; $H+$; Mg^{2+}
   b. Anions : Cl^- ;HCO_3^- ;HPO_4^{2-} ; SO_4^{2-}
B. Enzymes
   a. Pepsinogen
   b. Gelatinase – liquifies gelatin
   c. Gastric lipase
   d. Lysozymes – Bacteriocidal
   e. Urease – Hydrolyse urea to produce ammonia
C. Mucus
   a. Soluble mucus – pyloric and cardiac tubular glands
   b. Visible mucus – surface epithelium of gastric mucosa
D. Intrinsic factor – absorption of vitamin B12

MECHANISM OF HCL SECRETION
1. Secretion of H

a. Carbonic anhydrase helps in formation of H_2CO_3 from CO2 and H2O within the parietal cells
b. H2CO3 dissociates to form h and hco3
c. H is secreted into the lumen of canaliculi of parietal cell in exchange for K by H^+-K^+ ATPase pump
d. Hco3 produced are transported out through antiport in exchange with Cl at basolateral membrane.

2. Secretion of CL

   Because of high intracellular negativity, Cl^- is forced out through Cl^- channel on apical membrane.

   Cause of high intracellular negativity:
   1. Na^+-K^+ pump located at the basolateral membrane pumps out $3Na^+$ for $2k^+$
   2. K+ pumped in diffuses out through k+ channels present on basolateral as well as apical membrane.

REGULATION OF SECRETION OF GASTRIC JUICE

There are two regulatory mechanisms for regulation of secretion of gastric juice

1. Nervous mechanism –
   a. Local autonomic reflexes
   b. Impulses from the CNS → X nerve (vagal nerve)
2. Humoral mechanism
   a. Mediated by gastrin and histamine

PHASES OF GASTRIC JUICE SECRETION

Neural and hormonal mechanisms regulate the release of gastric juice. Stimulatory and inhibitory events occur in 3phases

1. Cephalic phase
2. Gastric phase
3. Intestinal phase

CEPHALIC PHASE	Gastric phase	Intestinal phase
• Psychic stimuli • Vagally mediated • 30-50% of total gastric juice secretion.	• Presence of food in the stomach • Local reflex and responses to gastrin • 50-60% of total gastric juice secretion	• Presence of food in the duodenum • Reflex and hormonal feedback • Very less secretion.

PEPTIC ULCER

It is the breakdown of the mucosal epithelium of stomach or duodenum

Causes –

1. Infection with a bacterium , helicobacter pylori. It stimulates H^+ secretion and inhibits HCO_3 secretion
2. Ethanol
3. Bile salts